

Title: Speed

Starter:

1. What is a resultant force?

The sum of all forces acting on an object

2. If the resultant force is 0N what would the motion of the object be?

It is stationary / moving at a constant speed

3. Define speed?

Distance travelled per unit time.

4. What unit can we use to measure speed?

km/h mph m/s

10:00



Success criteria

Lesson Title	Working towards	Working at	Working beyond
Recap: Forces & Motion (Y7)	I can identify different forces acting on an object.	I can describe what balanced and unbalanced forces are.	I can explain how unbalanced forces change an object's motion using force diagrams.
Speed	I can identify the correct units for speed, distance, and time.	I can use the speed = distance $\div$ time formula to calculate speed, distance or time.	I can solve speed questions using different units and rearranged formulas.
Describing Movement	I can read values from a distance-time graph.	I can describe motion based on the shape of the graph.	I can draw and explain distance-time graphs from written data or scenarios.
Turning Forces	I can give examples of things that turn when a force is applied.	I can use the moment formula to calculate turning forces.	I can explain how balanced and unbalanced moments affect turning.
Pressure between solids	I can state that heavier or larger objects create more pressure.	I can calculate pressure using the correct formula and units.	I can compare pressure in different situations using scientific reasoning.
Pressure in liquids and gases	I can describe how pressure changes with depth in liquids and altitude in gases.	I can explain what causes pressure in liquids and gases using particles.	I can calculate liquid pressure using the $h\rho g$ formula and explain examples.
Particles on the move	I can define diffusion.	I can describe how temperature and concentration affect diffusion.	I can explain diffusion using the particle model and apply it to real examples.

Speed

Keywords

- Constant
- Stationary

Learning objectives:

- Define speed and calculate average speed
- Recall and use correctly the units for speed

What 2 pieces of information would you need to know to calculate speed?



Think about Usain Bolt and how fast he ran? (units?)

What's the fastest land mammal?



Distance, time and speed

To work out the **speed** of an object you need to know:

- the **distance** travelled
- **Time** it took to travel that distance.



Usain Bolt

..... So his speed is equivalent to.....?!



Usain Bolt

- So his speed is equivalent to...

running the length of this lab in a second!!!

What's the fastest land mammal?
And how does a human compare?:



Think, Pair, Share

What is the fastest vehicle/ mode of transport you have ever been on?



Think, Pair, Share

In your opinion, what is the average speed of each of these modes of transport?

250 km/h to 260 km/h



Less than 10 km/h



285 km/h to 300 km/h



160 km/h to 185 km/h



840 km/h to 910 km/h

Think, Pair, Share

What unit is speed measured in?

250 km/h to 260 km/h



160 km/h to 185 km/h



Less than 10 km/h



285 km/h to 300 km/h



840 km/h to 910 km/h



There is an equation which links the **time it takes** to move somewhere, the **distance** the thing moves and the **speed** it is moving at.

For example, if a car moves **60 kilometres** in **one hour** we say it is moving at a speed of **60 kilometres per hour (60km/h)**

True or False?

- *If I travel 30 miles in an hour I'm going at the same speed as someone who travels 60 miles in two hours*

(Can you think what the equation might be?!)

True or False?

- *A car that passes the speed markings in two seconds is going faster than the car that passes them in one second*

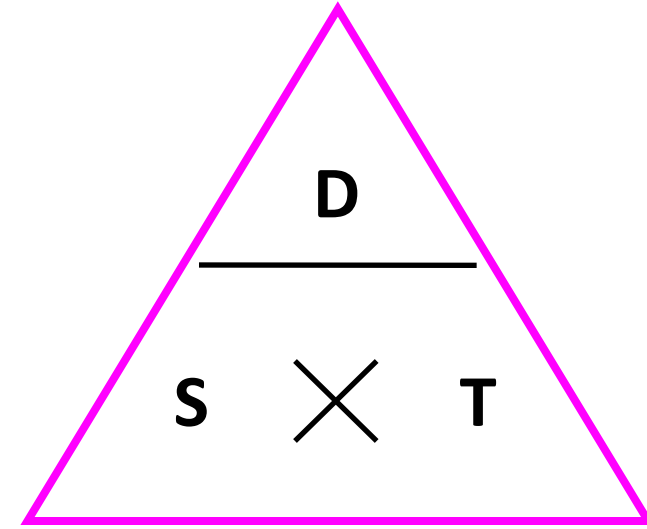
True or False?

- *If a car travels constantly and covers 1 mile in 1 hour, the car will have travelled 4 miles in 4 hours*
- *So what IS the equation?!*

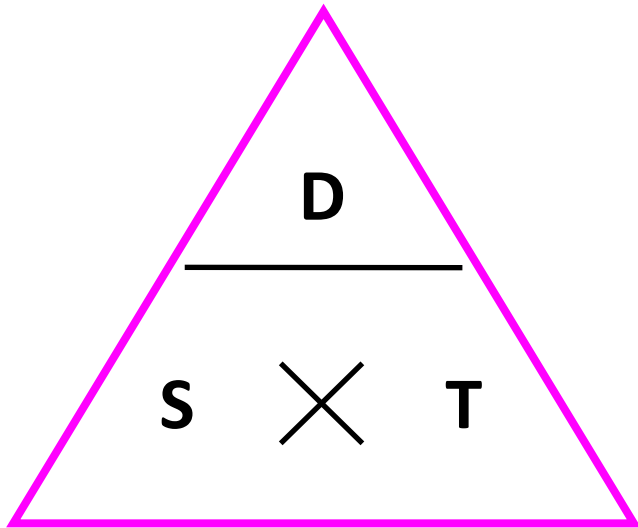
Distance, Speed and Time

$$\text{Speed} = \frac{\text{distance (in metres)}}{\text{time (in seconds)}}$$

m/s



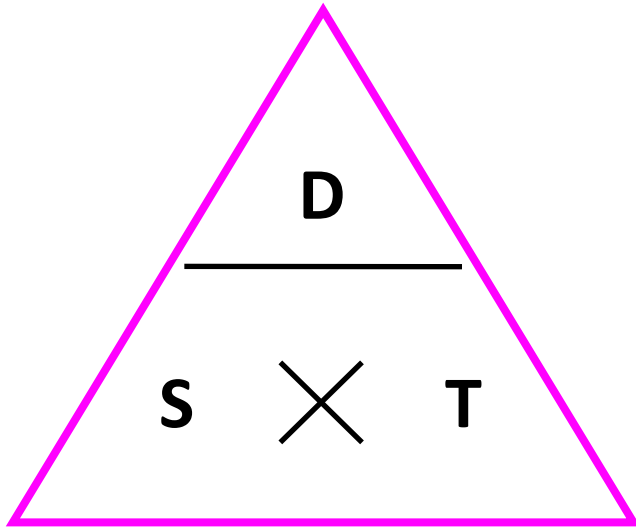
A coyote walks 320m in 80s. What is his speed?



$$\begin{aligned}s &= d/t \\ &= 320/80 \\ &= 4 \text{ m/s}\end{aligned}$$



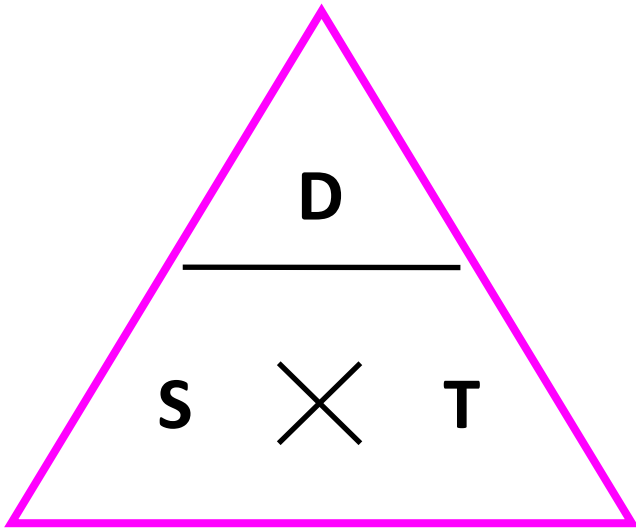
A badger walks 20m in 40 seconds.
What is its average speed in m/s?



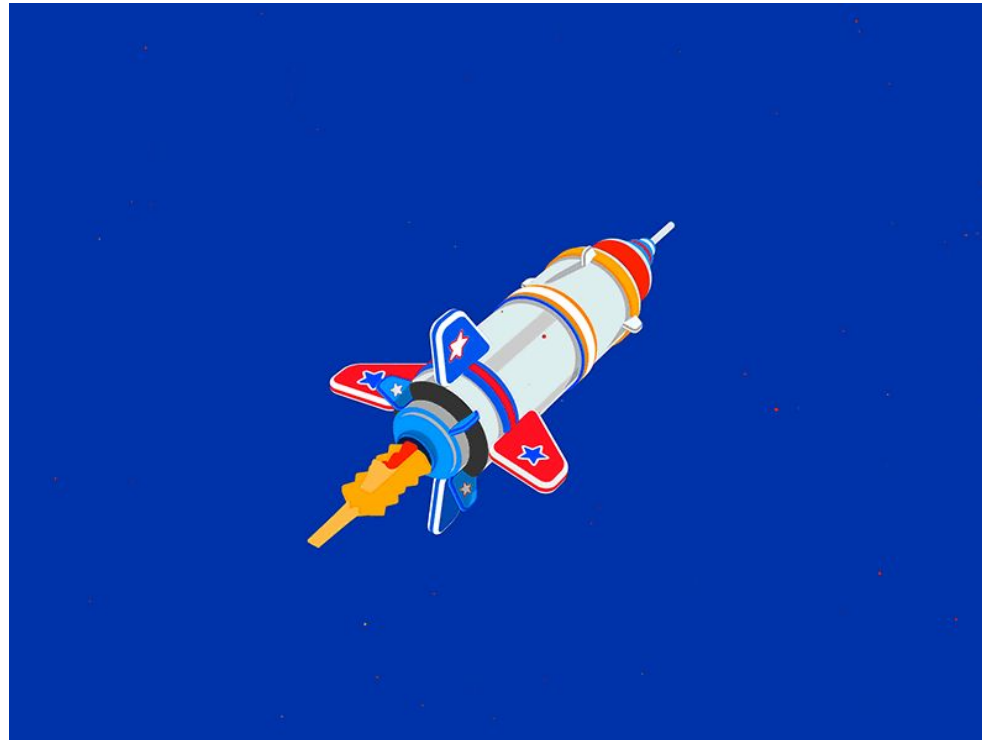
$$\begin{aligned}s &= d/t \\ &= 20/40 \\ &= 0.5 \text{ m/s}\end{aligned}$$



Then spaceship travels at 250m/s. How long does it take to travel 1500m?

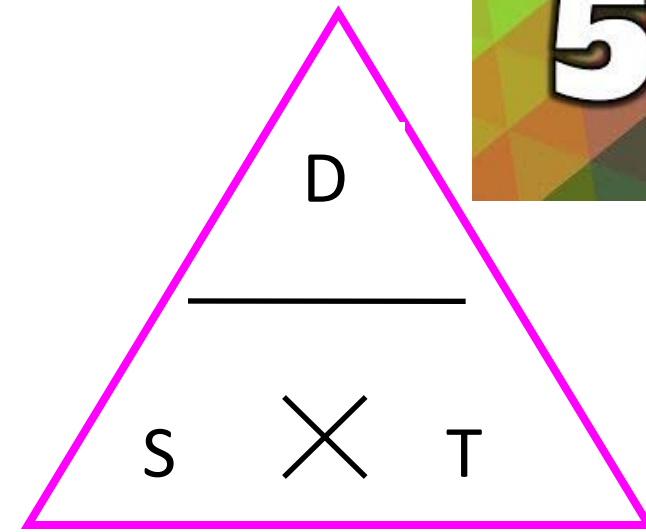


$$\begin{aligned} s &= d/t \\ t &= d/s \\ &= 6.0 \text{ m/s} \end{aligned}$$



Distance, Speed and Time

$$\text{Speed} = \frac{\text{distance (in metres)}}{\text{time (in seconds)}}$$



5:00

- 1) Dave walks 200 metres in 40 seconds. What is his speed?
- 2) Laura covers 2000m in 1,000 seconds. What is her speed?
- 3) How long would it take to run 100 metres if you run at 10m/s?
- 4) Steve travels at 50m/s for 20s. How far does he go?

Stretch: Susan drives her car at 85mph (about 40m/s). How long does it take her to drive 20km?

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Stretch: Susan drives her car at 85mph (about 40m/s). How long does it take her to drive 20km?

$$\begin{aligned} 1) \quad s &= d/t \\ &= 200 / 40 \\ &= \underline{5\text{m/s}} \end{aligned}$$

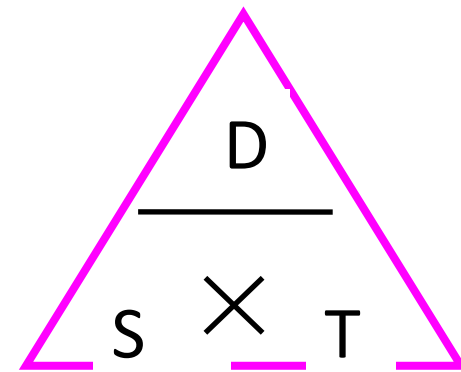
$$\begin{aligned} 2) \quad s &= 2000 / 1000 \\ &= \underline{2\text{m/s}} \end{aligned}$$

$$\begin{aligned} 3) \quad s &= d/t \\ t &= d/s \\ t &= 100/10 \\ &= \underline{10\text{s}} \end{aligned}$$

$$\begin{aligned} 4) \quad d &= s \times t \\ &= 50 \times 20 \\ &= \underline{1000\text{m}} \end{aligned}$$

$$\begin{aligned} 5) \quad 20\text{km in m?} \\ \underline{20000 \text{ m}} \end{aligned}$$

$$\begin{aligned} t &= d/s \\ &= 20000 / 40 \\ &= \underline{500\text{s}} \\ &= \underline{8.3 \text{ mins}} \end{aligned}$$



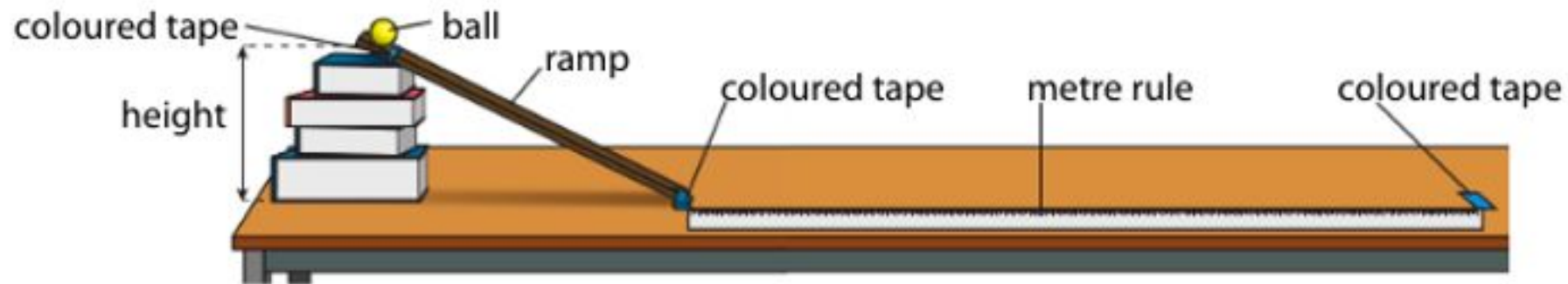
Learners book page 82-83

You will need:

- ramp (such as a plank of wood or thick card)
- tennis ball
- metre rule
- coloured tape
- books
- smooth level surface (such as a desk or the floor)
- stopwatch

Method

Set up the equipment as shown in the diagram.



Calculating speed

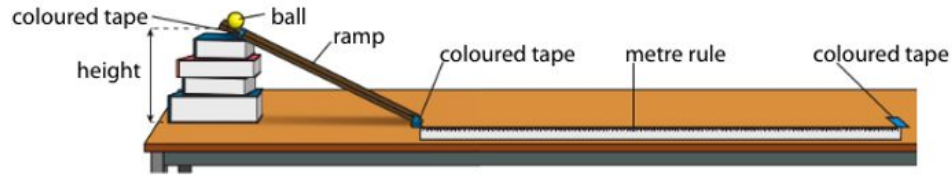
In this investigation, you will make measurements to calculate the speed of a tennis ball. Work in groups of three or four.

You will need:

- ramp (such as a plank of wood or thick card)
- tennis ball
- metre rule
- coloured tape
- books
- smooth level surface (such as a desk or the floor)
- stopwatch

Method

Set up the equipment as shown in the diagram.



- 1 Use coloured tape to fix the bottom of the ramp to the desk or floor.
- 2 Fix some coloured tape 1 m from the end of the ramp.
- 3 Fix some coloured tape near the top of the ramp to mark where you will release the ball.
- 4 Measure the height from the desk or floor up to the position where you will release the ball.
- 5 Release the ball and measure the time the ball takes to move between the two pieces of coloured tape on the desk or floor.
- 6 Repeat this two more times and calculate the average time to travel between the two pieces of tape.
- 7 Do this for a range of different heights.

Learners book page 82-83

Results

Independent variable =

Dependent variable =

Height of the ramp (cm)	Time taken (s)			Speed (m/s)
	1st measurement	2nd measurement	Mean	
5				
10				
15				
20				

Example Results

Independent variable = height of the ramp

Dependent variable = Time taken for the ball/cart to travel 1m

Height of the ramp (cm)	Time taken (s)			Speed (m/s)
	1st measurement	2nd measurement	Mean	
5	1.9	1.7		
10	1.5	1.3		
15	1.0	1.2		
20	0.5	0.7		

Example Results

Independent variable = height of the ramp

Dependent variable = Time taken for the ball/cart to travel 1m

Height of the ramp (cm)	Time taken (s)			Speed (m/s)
	1st measurement	2nd measurement	Mean	
5	1.9	1.7	$1.9 + 1.7 = 3.6$ $3.6 / 2 = 1.8$	$1 / 1.8 = 0.56$
10	1.5	1.3		
15	1.0	1.2		
20	0.5	0.7		

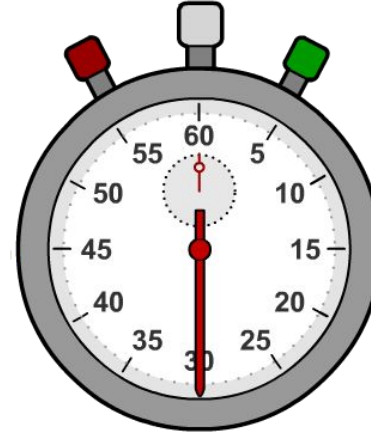
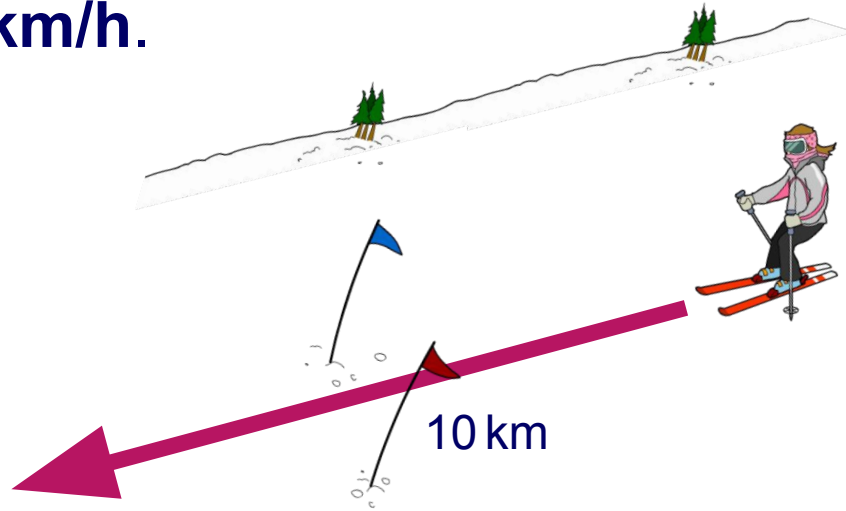
Results and questions

- 1 Record your results in a table.
- 2 Use your results to calculate the speed of the ball between the two pieces of tape. Add another column to your table, or draw a new table, to include the speed. Remember to put the unit of speed in the column header.
- 3 Plot a line graph of the results. Put *height* on the horizontal axis and *speed* on the vertical axis. Include the units on each axis.
- 4 What is:
 - a the independent variable in this investigation
 - b the dependent variable in this investigation?
- 5 State **two** variables that were controlled in this investigation.
- 6 Explain why each measurement is repeated. Give **two** reasons.
- 7 Describe the trend in your results.

Stretch

Speed calculation example

A girl takes 30 minutes to travel around a cross-country ski course: a distance of 10 km. Calculate her average speed in **km/h**.

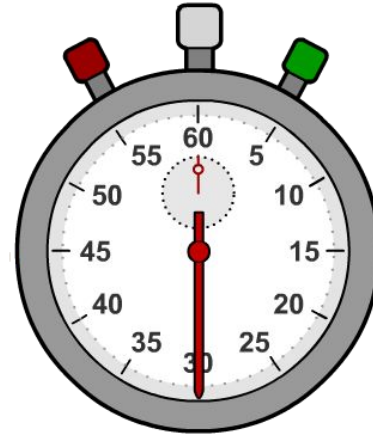
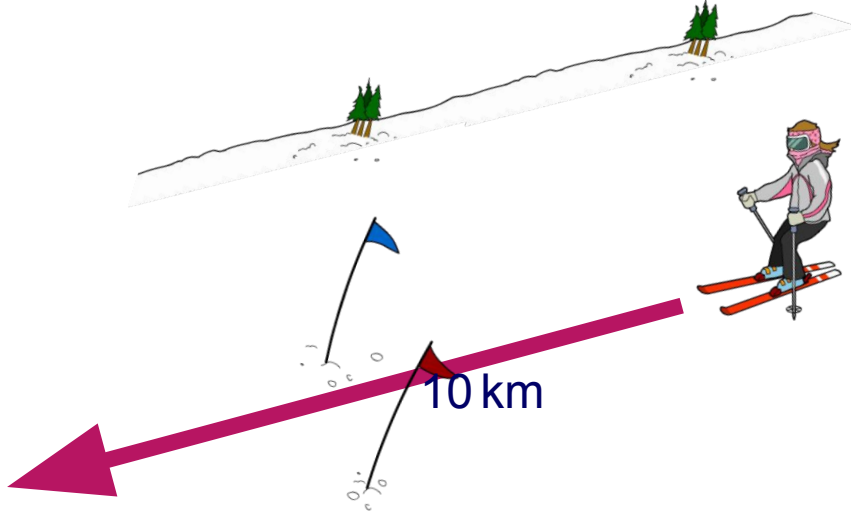


$$\begin{aligned}\text{average speed} &= \frac{\text{distance (in km)}}{\text{time (in h)}} \\ &= \frac{10 \text{ km}}{0.5 \text{ h}} = \mathbf{20 \text{ km/h}}\end{aligned}$$

Speed calculation example – units

Sometimes the units have to be changed in a speed calculation.

Calculate the average speed of the skier in **m/s**.



$$\begin{aligned}\text{average speed} &= \frac{\text{distance (in km)}}{\text{time (in h)}} \\ &= \frac{10 \text{ km}}{0.5 \text{ h}} = \mathbf{20 \text{ km/h}}\end{aligned}$$

$$\begin{aligned}\text{average speed} &= \frac{\text{distance (in m)}}{\text{time (in s)}} \\ &= \frac{10,000 \text{ m}}{1800 \text{ s}} = \mathbf{5.5 \text{ m/s}}\end{aligned}$$